USER INTERFACE DESIGN

EX No: 5 **240701463**

Task Analysis, User Flow Design, and Wireframe Development for an Online Book Store Application

---

AIM

To analyse and document the sequence of tasks performed by users in an online **book store application** and represent the workflow using user flow diagrams created in Lucidchart.

OBJECTIVE

• To understand the interaction between the user and the application during online book purchasing.
• To identify the steps involved in browsing books, selecting formats, adding items to the cart, and completing the checkout process.
• To represent the task flow visually using Lucidchart diagrams.
• To improve usability understanding through structured user flow documentation.

TASK ANALYSIS

The main tasks involved in an online book store application include:

1. Browsing books
2. Searching for a specific book
3. Selecting book formats
4. Adding books to the cart
5. Completing the checkout process

These tasks represent the primary user journey from entering the application to successfully placing an order.

USER FLOW ANALYSIS

1. Login / Register Flow

1. The user opens the application.
2. The system checks whether the user already has an account.

3. If the user is registered, they proceed with **Login**.
4. If the user is not registered, they choose **Register** and create a new account.
5. After successful login, the user is redirected to the home page.

---

2. Browsing Books

1. The user lands on the home screen.
2. The user selects a category such as **Fiction** or **Non-Fiction**.
3. The system displays the available books under the selected category.
4. The user browses through the list of books.
5. The user selects a book to view detailed information.

Flow Representation

Home Screen → Select Category → Browse Books → View Book Details

---

3. Checking Book Availability

1. The user views book details including title, author, price, and description.
2. The system checks whether the selected book is available.
3. If the book is not available, the user can add the book to the **Wishlist**.
4. If the book is available, the user proceeds to select the book format.

---

4. Selecting Book Format

1. The user selects the preferred format of the book.
2. Available formats may include:

• Paperback
• Hardcover
• eBook

3. After selecting the format, the user proceeds to add the book to the cart.

---

5. Adding Book to Cart

1. The user clicks **Add to Cart** to store the selected book.
2. The cart stores all selected items before checkout.

Flow Representation

Book Details → Select Format → Add to Cart

---

6. Checkout Process

1. The user proceeds to the payment stage.
2. The user selects a preferred payment method.

Available payment methods include:

• UPI
• Cash on Delivery (COD)
• Card Payment

3. If the payment fails, the user can retry the payment process.

---

7. Order Completion

1. After successful payment, the user enters **Shipping Details**.
2. The system processes the order.
3. The user receives **Order Confirmation** after the order is successfully placed.

Flow Representation

Cart → Payment → Shipping Details → Order Confirmation → End

PROCEDURE

Step 1: Create a New Diagram

1. Open Lucidchart.
2. Sign in or create a new account.
3. Click **Create New Document**.

Step 2: Choose Diagram Type

1. Select **Blank Flowchart Template**.

Step 3: Add Flowchart Shapes

Use different shapes to represent system actions.

• **Oval** → Start / End
• **Rectangle** → Process / Action
• **Diamond** → Decision
• **Arrow** → Flow direction

Step 4: Construct the User Flow Diagram

Develop the user flow diagram by identifying the sequence of interactions a user performs while purchasing a book through the application.

1. Begin the flowchart with the **Start** symbol representing the entry point of the system.
2. Add a **decision step** to determine whether the user is already registered in the application.
3. If the user is not registered, they are directed to the **registration process** to create a new account.
4. If the user is registered, they proceed to the **login process** to access the system.
5. After successful login, the user selects a **book category** such as Fiction or Non-Fiction.
6. The system displays a list of books, allowing the user to **browse the available books**.
7. The user selects a book and proceeds to **view detailed information** about the book.
8. The system checks whether the selected book is **available in stock**.
9. If the book is unavailable, the user may choose to add it to their **wishlist**.
10. If the book is available, the user proceeds to **select the preferred format** such as Paperback, Hardcover, or eBook.
11. The selected book is then **added to the shopping cart**.
12. The user proceeds to the **payment stage** and selects a preferred payment method such as UPI, Card, or Cash on Delivery.
13. Once the payment is successful, the user enters the **shipping details** for delivery.

14. Finally, the system displays the **order confirmation**, indicating that the purchase has been successfully completed.
15. The flow ends with the **End** symbol.

Step 5: Connect Shapes

Use arrows to connect the shapes and represent the direction of the process.

Step 6: Customize Diagram

- Use different colors for different types of steps.
- Group similar processes together.
- Ensure the decision points clearly show **Yes / No** paths.

Step 7: Save and Export

Save the diagram and export it as an image or PDF for documentation.

USER FLOW DIAGRAM

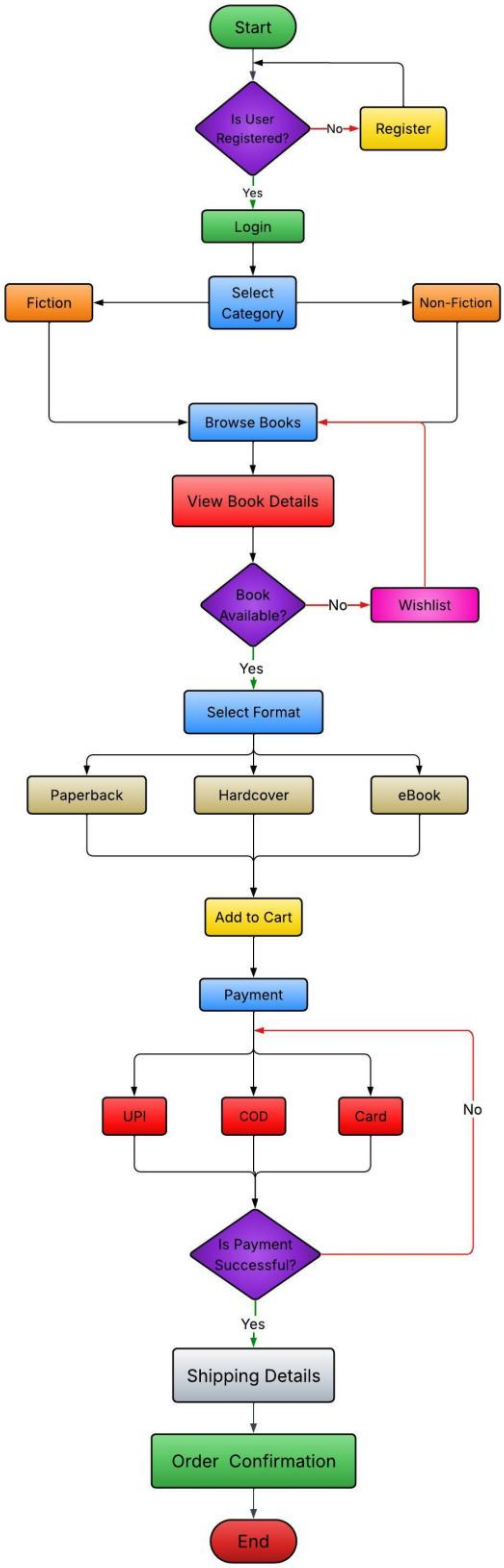


Figure 1: User Flow Diagram for Online Book Store Application

WIREFRAME DESIGN

Wireframes are simplified visual representations of a user interface that illustrate the layout and structure of an application. They focus on the arrangement of interface elements such as navigation bars, input fields, buttons, images, and content sections without emphasizing the final visual design or colors.

In this experiment, wireframes were created for the **Online Book Store Application** to represent the main screens involved in the user interaction process. These wireframes help visualize how users interact with the system while browsing books, selecting formats, adding items to the cart, and completing the checkout process.

The wireframes were designed to maintain a clear and consistent layout suitable for a mobile application interface. Each screen represents a different stage of the application workflow and follows the user flow diagram created earlier using Lucidchart.

The wireframes illustrate the structure and navigation flow of the application, ensuring that the user can easily move between screens while performing different tasks.

WIREFRAME SCREENS DESCRIPTION

The following wireframes were designed for the Online Book Store Application to represent different stages of the user journey.

Login / Registration Screen

The login screen allows users to access the application by entering their login credentials. Users who do not have an account can register by selecting the sign-up option. This screen typically contains fields for entering email or username, password, and login buttons.

Home Screen

The home screen acts as the main dashboard of the application. It provides access to book categories, search functionality, and recommended books. From this screen, users can easily navigate to different sections of the application.

Browse Books Screen

This screen allows users to browse available books in selected categories such as Fiction and Non-Fiction. Books are displayed in an organized layout where users can scroll through the list and select a book of interest.

Book Details Screen

The book details screen provides detailed information about the selected book. This includes the book title, author name, description, price, and rating. Users can also choose their preferred format such as Paperback, Hardcover, or eBook.

Wishlist Screen

The wishlist screen stores books that users want to save for later purchase. Users can review the saved books and move them to the cart when they are ready to buy them.

Shopping Cart Screen

The shopping cart screen displays all the books selected by the user. It provides options to modify quantities, remove items, and view the total price before proceeding to the payment process.

Payment Screen

The payment screen allows users to choose their preferred payment method. The application supports multiple payment options such as UPI, Cash on Delivery (COD), and card payments.

Shipping Details Screen

In this screen, users provide the required delivery information including name, address, contact number, city, and postal code to complete the order process.

Order Confirmation Screen

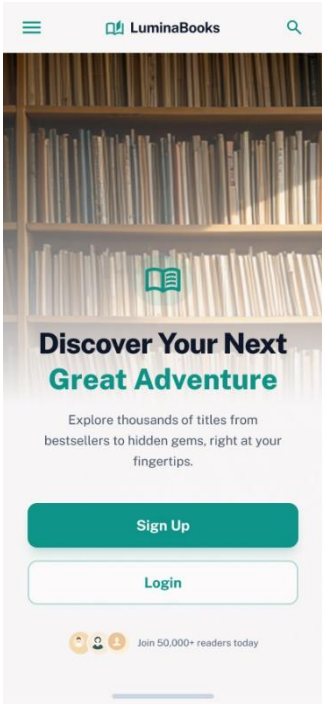
The order confirmation screen displays a confirmation message indicating that the purchase has been successfully completed. It may also include order details and estimated delivery information.

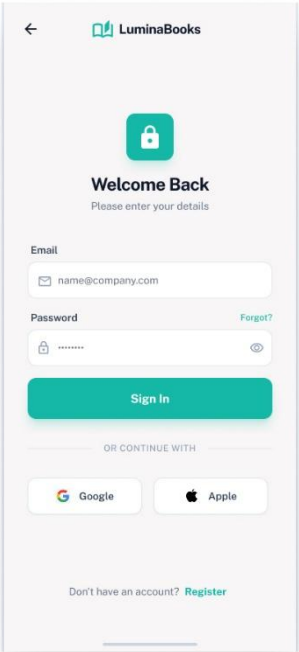
WIREFRAME IMPLEMENTATION

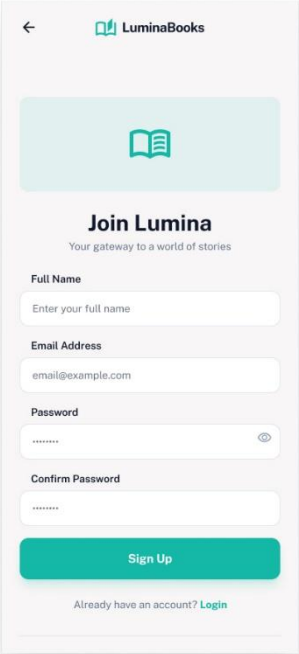
The wireframes were created using digital design tools to represent the structure of the application screens. These wireframes provide a visual understanding of how the user interface elements are arranged and how users interact with the application at each stage.

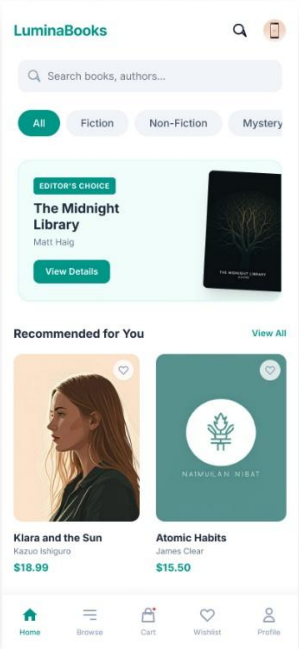
The wireframe design helps designers and developers evaluate usability, identify navigation flow, and improve the overall structure of the application before implementing the final interface design.

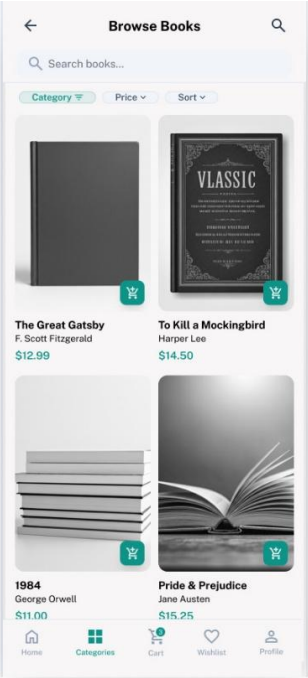
WIRE FRAME OUTPUT

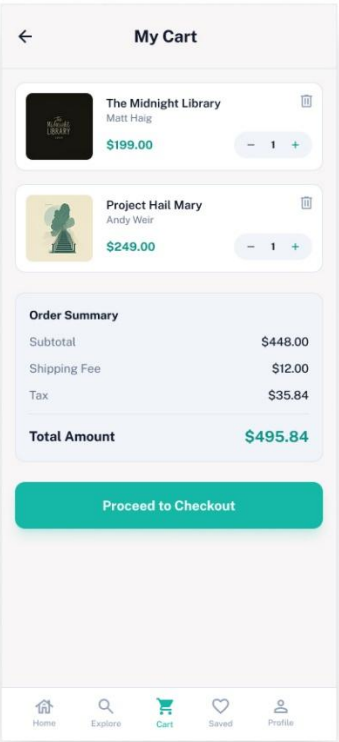


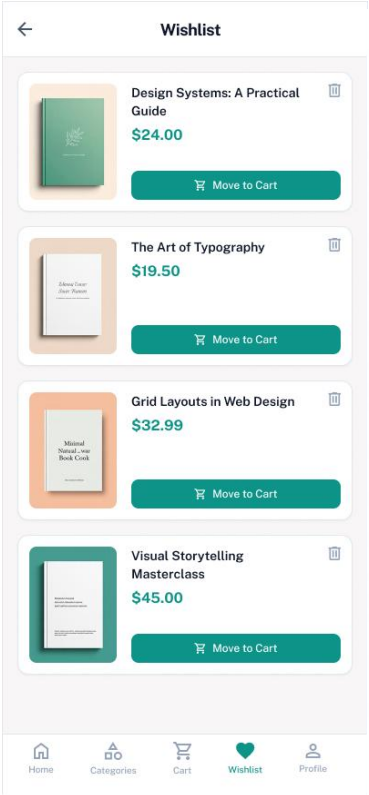


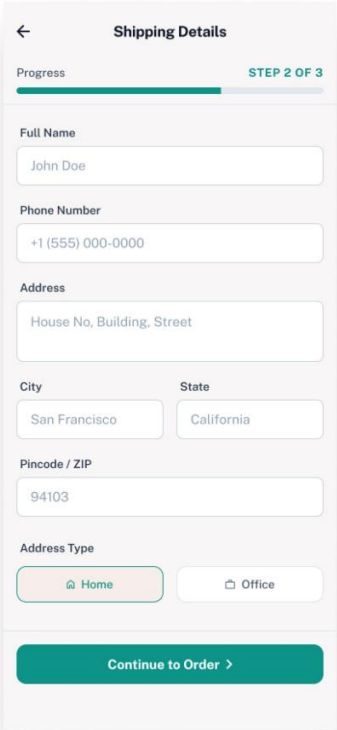


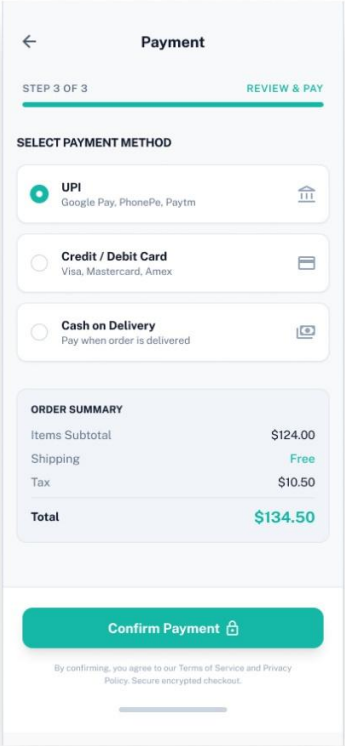


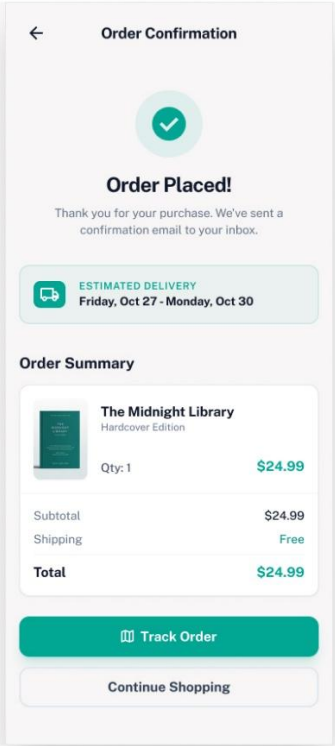



My Cart
The Midnight Library
Matt Haig
$199.00
1
Project Hail Mary
Andy Weir
$249.00
1
Order Summary
Subtotal $448.00
Shipping Fee $12.00
Tax $35.84
Total Amount $495.84
Proceed to Checkout
Home
Explore
Cart
Saved
Profile


Wishlist
Design Systems: A Practical Guide
$24.00
Move to Cart
The Art of Typography
$19.50
Move to Cart
Grid Layouts in Web Design
$32.99
Move to Cart
Visual Storytelling Masterclass
$45.00
Move to Cart
Home
Categories
Cart
Wishlist
Profile


Shipping Details
Progress
STEP 2 OF 3
Full Name
John Doe
Phone Number
+1 (555) 000-0000
Address
House No, Building, Street
City
San Francisco
State
California
Pincode / ZIP
94103
Address Type
Home
Office
Continue to Order


Payment
STEP 3 OF 3
REVIEW & PAY
SELECT PAYMENT METHOD
UPI
Google Pay, PhonePe, Paytm
Credit / Debit Card
Visa, Mastercard, Amex
Cash on Delivery
Pay when order is delivered
ORDER SUMMARY
Items Subtotal $124.00
Shipping Free
Tax $10.50
Total $134.50
Confirm Payment
By confirming, you agree to our Terms of Service and Privacy Policy. Secure encrypted checkout.


Order Confirmation
Order Placed!
Thank you for your purchase. We've sent a confirmation email to your inbox.
ESTIMATED DELIVERY
Friday, Oct 27 - Monday, Oct 30
Order Summary
The Midnight Library
Hardcover Edition
Qty: 1
$24.99
Subtotal $24.99
Shipping Free
Total $24.99
Track Order
Continue Shopping

RESULT

The wireframes for the **Online Book Store Application** were successfully designed to represent the layout and structure of the user interface. These wireframes demonstrate how users navigate through different screens including login, browsing books, viewing book details, adding books to the cart, selecting payment methods, and confirming orders.

The combination of the **Lucidchart user flow diagram and the application wireframes** helped in understanding the complete interaction flow of the system. This experiment improved the understanding of user interface design principles and demonstrated how wireframes can be used to visualize application functionality before development.